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Research Interests

Gaze-based interaction in AR/VR; gaze distance and depth ambiguity in near-eye target selection; intent understanding from gaze, distinguishing selection, exploration, and confirmation; eye–hand coordination and its computational (resource-rational) modeling; head-mounted eye tracking, gaze calibration, and implicit recalibration from natural behavior; target selection and multimodal (gaze + hand / head / foot) input.

Education

Southeast University, Nanjing, China Sep 2024 – present (expected Jun 2028)
PhD in Mechanical Engineering (Professional Doctorate) *Advisor: Prof. Yafeng Niu*

Southeast University, Nanjing, China Sep 2021 – Jun 2024
MS in Design Science *Outstanding Master's Thesis Award, Southeast University (2025)*

China Agricultural University, Beijing, China Aug 2017 – Jun 2021
BS in Industrial Design; second degree in Data Science and Big Data Technology *GPA 3.9/4.0, ranked 1/36*

Research Experience

Head-mounted binocular eye-tracking platform 2025 – present
Lead developer (hardware and algorithms)

- Built a low-cost head-mounted binocular eye tracker end to end: 3D-printed frame, near-infrared eye cameras, scene camera, and real-time pipeline.
- Developed NIR pupil detection, gaze calibration (including mirror-based camera extrinsic calibration), binocular vergence depth, and multi-camera data collection tools.
- Using the platform to study eye–hand coordination and model-driven implicit calibration (PhD thesis).

Gaze interaction techniques for AR/VR 2021 – present
First-author and co-author studies with Prof. Yafeng Niu's group

- Designed and evaluated gaze, blink, smooth-pursuit, gaze–head, gaze–foot, and gaze–hand selection techniques through controlled user studies (HoloLens 2, Unity, Tobii, Pupil Labs).
- Unobtrusive gaze calibration from mouse dragging (DragCal), with per-user eye–hand lag analysis.

Visual–haptic interaction mechanisms in gaze-controlled systems 2023 – Jun 2026 (completed)
Technical lead, Ministry of Education Joint Fund project

- Led the technical roadmap, experimental design, and test-system implementation; the project passed final acceptance testing in 2026.

Publications

Y.-Y. **Wang** = me; * = equal contribution; † = corresponding author.

Journal articles

[J1] Q. Xiao*, **Y.-Y. Wang***, Q. Ma*, et al., T.-J. Cui, D.-P. Tsai. Gaze-driven topological phase-transition metasurfaces. *Opto-Electronic Advances*, 2026. [Accepted]

[J2] **Y.-Y. Wang**, W.-C. Huang, L.-H. Fan, Z.-T. Wang, Y.-F. Niu. SPMark: Improving small-target text selection with smooth-pursuit gaze interaction in hybrid paper–digital environments. *Virtual Reality* (Springer), 2026. doi:10.1007/s10055-026-01497-x

[J3] J.-Z. Tian, **Y.-Y. Wang**, J.-C. Wu, C. Liu, Y.-F. Niu. Exploring coordination in gaze–hand cascaded interaction: Designing robust solutions for low-precision eye tracking. *Advanced Engineering Informatics*, 74, 104766, 2026.

[J4] W.-C. Huang, Z.-H. Song, **Y.-Y. Wang**, L.-H. Fan, Y.-F. Niu. Take-A-Look: Enabling intuitive distant target interactions in industrial virtual reality applications via natural gestures. *Computers in Industry*, 181, 104545, 2026.

[J5] Y.-F. Niu, Y. Feng, X. Li, W.-C. Huang, **Y.-Y. Wang**, L.-H. Fan. Visual perception and user experience optimization: Exploring interaction design through static-triggered smooth pursuit systems. *Displays*, 90, 103121, 2025.

[J6] W.-C. Huang, **Y.-Y. Wang**, L.-H. Fan, S.-T. Yang, Y.-F. Niu. Enhancing gaze interaction performance: Design and optimization of perspective-driven fisheye view. *Advanced Engineering Informatics*, 65, 103365, 2025.

[J7] Y.-F. Niu, **Y.-Y. Wang**†, W.-C. Huang, L.-H. Fan. Research status and progress of gaze interaction paradigm design based on design ergonomics (in Chinese). *Packaging Engineering*, 46(4), 20–35, 2025.

[J8] Y.-F. Niu, R. Chen, **Y.-Y. Wang**, X.-Y. Yao, Y. Feng. Research on click enhancement strategy of hand–eye dual-channel human–computer interaction system: Trade-off between sensing area and area cursor. *Advanced Engineering Informatics*, 62, 102880, 2024.

[J9] **Y.-Y. Wang**, J.-Z. Tian, L. Xiao, J.-X. He, Y.-F. Niu. Research on a spatial–temporal characterisation of blink-triggered eye control interactions. *Advanced Engineering Informatics*, 59, 102297, 2024.

[J10] **Y.-Y. Wang**, W. Cheng, J.-Z. Tian, W.-C. Huang, Y.-F. Niu, W.-J. Yang, C.-Q. Xue. Design recommendations of target size and tracking speed under circular and square trajectories for smooth pursuit with Euclidean algorithm in eye-control system. *Displays*, 81, 102608, 2024.

Manuscripts under review

[M1] **Y.-Y. Wang**, J.-X. He, J.-Z. Tian, W.-C. Huang, L.-H. Fan, Z.-T. Wang, R. Chen, Y.-F. Niu. Gaze2Foot: Foot primitives and stabilization mechanisms for gaze–foot target selection across postures. *ACM Transactions on Computer-Human Interaction.* [Under review]

[M2] **Y.-Y. Wang**, H.-B. Chen, W.-C. Huang, L.-H. Fan, Z.-T. Wang, Y.-F. Niu. Gaze localizes, the head confirms: Staged head–eye coordination for hands-free selection in dense AR. *International Journal of Human–Computer Interaction.* [Under review]

[M3] **Y.-Y. Wang**, J.-L. Cai, W.-C. Huang, L.-H. Fan, Z.-T. Wang, Z.-X. Yang, Y.-F. Niu. DragCal: Unobtrusive gaze calibration from mouse dragging. *International Journal of Human–Computer Interaction.* [Submitted]

[M4] **Y.-Y. Wang**, J.-S. Lv, W.-C. Huang, L.-H. Fan, Z.-T. Wang, Z.-X. Yang, Y.-F. Niu. Spatiotemporal evidence fusion and state-space analysis for multi-target gaze interaction: Coordinating trajectory geometry, spatial layout, and temporal confirmation. *Reliability Engineering & System Safety.* [Under review]

Conference papers (co-author)

[C1] H.-B. Chen, W.-C. Huang, **Y.-Y. Wang**, J. Liu, Y.-F. Niu. The design and experimental study of eye–hand integrated dual-channel interaction strategies. *IEEE IEEM*, 2024.

[C2] M.-J. Li, W.-C. Huang, **Y.-Y. Wang**, Z.-J. Han, Y.-F. Niu. Does increasing takeover time budget improve driver takeover performance in different hazard visibility scenarios? *IEEE IEEM*, 2024.

[C3] M.-L. Wu, Y.-N. Li, **Y.-Y. Wang**, J.-Z. Tian, W.-J. Yang, Y.-F. Niu. The effectiveness of audible alarm types and presentation rates on pilot performance in beyond visual range combat scenarios. *IEEE SMC*, 2023.

Plus 8 further co-authored papers at IEEE IHMSC (2025, ×4), IEEE IEEM (2023), M2VIP (2023), CIE50 (2023), and AHFE IHSI (2023); full list at george-wyy.github.io.

Book chapter

- Chapter 1 and technical editing, in Y.-F. Niu, *Eye-Control Interaction Paradigm Design* (in Chinese). Science Press, 2025. ISBN 978-7-03-083699-1.

Patent Applications (Chinese invention patents, pending)

Applicant: Southeast University. All applications published; inventor rank in brackets.

- Implicit adaptive gaze calibration for near-eye AR devices based on hand–eye coordination priors. CN122431533A, 2026. [2nd]
- Gaze-depth ambiguity resolution for near-eye AR devices. CN122369009A, 2026. [2nd]
- Multimodal LLM interaction disambiguation based on gaze-intent pattern classification. CN122286538A, 2026. [2nd]
- Cross-modal perceptual-confirmation interaction based on tactile motion illusion. CN122526419A, 2026. [2nd]
- Hybrid gaze–smooth-pursuit eye-touch small-target selection method and device. CN119045661A, 2024. [2nd]
- Smooth-pursuit-based hybrid digital-paper text annotation system and method. CN119045660A, 2024. [2nd]
- Error-reverse-guided gaze interaction method and system. CN121349306A, 2026. [3rd]
- DMA-based multi-channel ultrasonic transducer phased-array driving method. CN121763911A, 2026. [4th]

One further application as 2nd inventor accepted in Oct 2026, pending publication.

Earlier: eye-movement password input (CN116755544A, 2023, with AVIC); three agricultural-machinery patents from undergraduate work (one invention, two utility models; granted).

Teaching, Mentoring, and Service

- **Guest instructor**, eye-control interaction module, *Intelligent Human–Computer Interaction*, Southeast University, 2026 (lectures, demos, and hands-on lab sessions).
- **Mentoring**: guided junior master's students in the lab on study design and paper writing.
- **Reviewer**: *Displays* (Elsevier), 2025–2026; *International Journal of Industrial Ergonomics*, 2026; *International Journal of Human–Computer Interaction*, 2026.
- **Presentation**: Doctoral and Master's Forum on Design Ergonomics, Chinese Ergonomics Society, Sep 2026.

Honors and Awards

Outstanding Master's Thesis, Southeast University (2025) · National Scholarship, China (2019, 2020)

Skills

- **Eye tracking and XR**: Tobii, Pupil Labs, custom head-mounted trackers; HoloLens 2, Unity, MRTK3.
- **Programming and analysis**: Python (PyTorch, OpenCV), C#, R; experimental design, statistical modeling, usability evaluation.

- **Hardware**: camera calibration, NIR illumination, 3D printing and CAD for wearable prototypes.
- **Languages**: Chinese (native), English (working proficiency).